Continuity logic

Document continuity — where one document ends and the next begins

What the code does, with every number and the file that holds it. The method and its reasoning come from *How we decide where one document ends and the next begins* (Document_Splitting.md, 2026-10-20); this file describes it as built. Paths are relative to `core-pipeline/`.

Engine	`stages/lib/continuity/engine.py` (`assign`, `score_gap`, `_final_pass`)
Measurements	`stages/lib/continuity/signals.py` (furniture, tail / head blocks, TF-IDF, the curves)
Stage	`stages/lib/continuity/stage.py` — `continuity`, seq 87, after `page_subtype` and `dos_extract`, before `imaging_final`
Settings	`stages/lib/keyword-canon/continuity_canon.json`, version 4 (reloads on change)
Inputs	`staging/extraction.json` first (§3), then the DB (DOS, page type, blank / junk) and the OCR JSON
Writes	`page_continuity_results` (the original columns plus `confidence`, one row per content or junk page), `<chart>/staging/continuity.json` (every score of every gap), `imaging/<chart>_continuity.csv` (every page)
Database	one new column, `confidence` (0–1); on an existing database `schema/changes_2026-10-19.sql` §8 widens `decided_by` and §9 adds `confidence`

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1. Why it matters

The pages of a chart arrive in order with nothing saying where one document stops. Four later decisions depend on the grouping:

- **Date of service.** A labelled date is on about 18% of pages; the rest inherit their document's.
- **Page type.** A lab table inside a progress note is part of the note.
- **Codability.** The same questionnaire is codeable inside a note and not on its own.
- **Encounter type.** Decided once per document (`encounter` stage, seq 90).

The two mistakes are not equal. Joining two documents merges two visits, and nothing downstream can see it. Splitting one leaves a fragment a reviewer notices. So **when the evidence is balanced, the gap joins**, and a later pass can still split.

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2. Gaps, not pages

Between any two consecutive pages there is a **gap**, and the gap is what is decided. It gets one **break score**: a plain sum of points, positive toward a break, negative toward joined.

- **Break Before** of a page = the score of the gap before it.
- **Break After** of a page = the score of the gap after it.
- Break After of page 4 and Break Before of page 5 are **the same number**, so the two pages can never disagree.
- The first page of the chart always starts a document; the last always ends one.

Every gap draws on three kinds of evidence, kept as separate **sub-scores**:

Kind	Read from	Example
ending	the earlier page alone	it finishes with a signature
opening	the later page alone	it carries a title
pair	the two compared	the same header line on both

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3. Where each fact comes from: `extraction.json` first

The key/value extraction runs once after OCR and stages what it found per page in `<chart>/staging/extraction.json`. Continuity reads it first (`stage.py`):

Fact	From the staging	Fallback
How the date was found (DOS trust)	the tier of the staged `dos` row that read the page's Extracted DOS (best tier wins)	`dos_trust.default` (3)
Printed page number	the selected `page_no` row (`page_no`, `page_total`)	`additional_page_details`, then "page N of M" in the header / footer bands
Signed anywhere	any accepted `electronic_signature` row	`provider_signature.csv` `signature_present = y`, or a signature phrase in the text
Signed in the closing lines	an accepted `electronic_signature` row in the `footer` region	a signature phrase in the page's last third of lines
Visit number	accepted `member_id` rows whose key is an encounter-type ID (FIN, CSN, account, visit — `member_id/id_types.json`)	`Visit / Encounter / FIN / CSN / Account #` in the text
Section headings	accepted `heading_heron` rows of kind `section`	the OCR JSON's matched `section_headers` (both are used)

The rest comes from the database and the OCR files:

Fact	From
Date of service	`dos_extraction_results` (the DOS stage's Extracted DOS)
Page type, full path, confidence, title	`page_classification` (axes: family = L1, names = the full path; `axes.title_of`)
Blank / junk / duplicate, junk subtype and confidence	`v_page_blank_junk_final`
Line positions	Final 2 JSON line polygons, else Final 1 words grouped into lines, else text lines

Header band: a line whose top is above **10%** of the page; footer band: bottom below **92%** (`bands`).

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4. Every signal and its points

Points are **1, 2, 3 or 5** only (`points` in the canon): 5 decisive, 3 strong, 2 moderate, 1 weak. + pushes toward break, – toward joined.

4.1 Decisive — 5

Signal	Points	When
`different_dos`	**+5 / +3 / +2**	the page's date differs from the open document's date
`same_dos`	**–5 / –3 / –2**	the page's date equals the open document's
`pagination_follows`	**–5**	"page 3 of 5" followed by "page 4 of 5"

Date points depend on how the date was found (`dos_trust`); the lower of the two dates' trust counts:

DOS tier	Points
`service`, `admit`, `discharge` (a proper label)	**5**, decisive
`weak`, `received`, `printed`, `collected` (a keyword near the edge; lab collection dates kept off the decisive tier)	3
`keyless`	2
not explained by the staging	3

The open document's date is its first dated content page. A junk segment owns no date, so the last document's date stays open across it.

Page numbers only ever join. "Page 1 of 6" is not used to split: OCR misreads a 7 as a 1, and a wrongly split document is very hard to find later.

4.2 Strong — 3

Signal	Side	Points	Measured
`signed_in_tail`	ending	+3	difference 21
`signed_anywhere`	ending	+2	difference 26 (fires with the above: a signed closing page totals +5)
`last_numbered_page`	ending	+3	“page N of N”
`has_title`	opening	+3	**assumed — check first**: section headings taken for titles
`page_type_change`	pair	+3 × (1 – same-type probability) × confidence	assumed (§5.2)
`junk_run`	pair	+3 (a run of 2) / +5 (3 or more) × junk confidence	assumed (§6)
`says_continued`	ending	–3	“continued”, “cont’d”, “(over)”, “see next page” at the end
`sections_forward`	pair	–3 × how far forward	difference 27, no false positives
`same_visit_number`	pair	–3	difference 8, no false positives

4.3 Moderate — 2

Signal	Side	Points	Measured
`page_fill`	ending	–2 .. +2, the curve in §5.1	join side difference 34, the strongest measured
`text_start`	opening	–1 .. +2, §5.4	**unmeasured: recorded only**
`same_header_line`	pair	–2	difference 31 (normalised matching should beat it)
`wording_char`	pair	–2 × similarity	difference 27, after removing furniture (§5.3)
`same_junk_kind`	pair	–2	both junk, the same subtype

4.4 Weak — 1

Signal	Side	Points	Note
`blank_between`	pair	+1	a blank page in the gap
`same_footer_line`	pair	−1	difference 8: one footer template is printed across unrelated documents
`wording_word`	pair	−1 × similarity	not independent of `wording_char`, so it carries less
`cut_mid_sentence`	ending	−1	only when `wording_char` fires: worthless alone (difference 3)
`starts_mid_thought`	opening	−1	only when `wording_char` fires (difference 5 alone)
`same_page_type`	pair	−1	the full type path, not just “both Progress Notes”

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5. The scaled signals

5.1 Where the text ends (`page_fill`)

From the bottom of the earlier page’s last body line (lines starting in the footer band do not count), as a fraction of the page height:

Text ends at	Value	Reading
≥ 92%	**−2**	the page ran out of room: continues
80% – 92%	−2 .. 0	tapering off
65% – 80%	0 .. +2	rising into the band
50% – 65%	**+2**	finished, with room for a signature
35% – 50%	+2 .. 0	falling away
< 35%	**0**	too little on the page to judge (a spillover tail or a failed print)

The peak is worth only +2, so a form page ending in the band can be outvoted. With a signature in the closing lines it adds up to near-decisive, which is right.

Without line positions a rough proxy is used: −2 when the page has ≥ 28 lines or ≥ 1,800 characters, else nothing. The positive band cannot be read this way. When the proxy decides a gap, the gap is flagged.

5.2 How much the page type changed (`page_type_change`)

`(1 − same-type probability) × the classifier's confidence × 3`. The same-type probability is the confidence when the two families agree, one minus it when they differ. Below 0.05 it is not counted.

Previous	This page	Confidence	Points
Progress Note	Lab	90%	+2.4
Progress Note	Lab	60%	+1.1
Progress Note	Progress Note	95%	+0.1

5.3 Wording across the gap (`wording_char`, `wording_word`)

1. **Remove the furniture.** Any line of ≥ 12 characters printed on both pages (compared normalised, §7) is stripped: banner, letterhead, footer template. Without this the signal measures nothing (difference 1). The removed lines are what `same_header_line` and `same_footer_line` score instead.
2. **Take the ends by line.** The earlier page's **last two body lines** by position, extended back until 20 words; the later page's first two, forward.
3. **Compare.** Character 3–5 grams (survives OCR damage) and word 1–2 grams, TF-IDF fitted on every block of the chart, so words every page shares are discounted. Similarity = cosine / 0.25 (characters) or / 0.15 (words), clipped to 1; below 0.05 it is not counted.

5.4 Where the next page's text begins (`text_start`)

How far below the header band's lines the later page's first body line starts: ≤ 3% → −1, 3–12% → −1 .. 0, 12–25% → 0 .. +2, > 25% → +2. **Unmeasured, so it is computed and stored (`recorded.text_start`) but not added to the score**, until `text_start.apply` is set after a run has measured it.

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6. Blank pages, junk and duplicates

Kind	Treatment
Blank	Transparent. The gap is scored between the page before and the page after, plus `blank_between` +1. The blank page has no scores of its own and is shown as **"Blank (in Doc 3)"** when that gap joined, **"Blank"** between documents.
Single junk page	Transparent and adds nothing: the gap is scored between the content pages either side. It belongs to the document before it (an invoice stapled to the back of a note), keeps its own junk page type, and gets no document number for Final values.
Run of two or more junk pages	Scored like content and broken off at both ends by `junk_run`: +3 for two pages, +5 for three or more, × the run's mean junk confidence. Three or more with a scaled value ≥ **4.5** is decisive. The run becomes its own **junk group**, numbered with the documents: **"New (Junk 4)" … "Last (Junk 4)"**. Inside it, `same_junk_kind` −2 holds it together. Blanks and duplicates between junk pages do not break a run. A junk group is no visit: its **DOS and encounter type read "Skipped"**.
Duplicate	Takes its original's document and adds no gap.

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7. Header and footer lines are matched normalised

Every run of digits and every date is replaced by a placeholder before an **exact** comparison, so a page number or a print time cannot make an identical banner differ:

```
raw         Smith, John | MRN 004812 | Printed 03/28/24 | Page 3 of 7
normalised  smith, john | mrn # | printed <date> | page # of #
```

Fuzzy matching was measured and is noise (difference –4 for headers, 4 for footers), so the match stays exact.

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8. The decision

8.1 Every gap is decided

Score	Decision
> 0	**break**
< 0	**joined**
= 0	**joined, flagged** (`decided_by tie`)

There is no “don’t know”: every page needs a document to inherit its date, type and encounter from.

8.2 Decisive signals

- **A 5-point signal settles the gap** when no other 5 opposes it: the gap takes its sign whatever the rest says (`decided_by decisive`, confidence high). The other signals are still computed and stored.
- **Two decisive signals disagree** (page numbers run on, but the dates differ): **the different date wins** and the gap breaks, at medium confidence, flagged (`decisive_conflict`). A date is a fact about the care; a page number is a fact about the printing.

8.3 Confidence

Field	How
`confidence_level`	`|score|` ≥ 5 high, 2–4 medium, 0–1 low; decisive high; decisive conflict medium
`confidence` (0–1)	`P(break) = 1 / (1 + exp(−(score − b) / a))`, confidence = `|2P − 1|`; × **0.6** when the three sub-scores point opposite ways; at least 0.9 for a decisive gap. **a = 2, b = 0 until fitted** on the labelled gaps

A low-confidence side of a page adds a **“?”** to its label: `Continue? (Doc 3)`.

8.4 When the sub-scores disagree sharply

Two patterns, at ±4 (`conflict_rules.at`):

Pattern	Sub-scores	Usual cause	Right answer
A finished, but still matching	ending ≥ +4 and pair ≤ –4	a signed note followed by an addendum from the same visit	break
B new-looking page, continuing content	opening ≥ +4 and pair ≤ –4	a section heading taken for a title	join; record it as a title misfire

Either one **drops the confidence one level and flags the gap**. The answer itself is changed only when `conflict_rules.apply` is true. It is **false** for now, because the document says to count how often each pattern appears on a run before wiring them in. The pattern is stored as `recorded.conflict`.

8.5 What is flagged (`review_required`)

- low confidence (`confidence < 0.25`, or level low);
- a score of exactly 0;
- two decisive signals disagreeing;
- conflict pattern A or B;
- the page-fill proxy decided the gap;
- a final-pass split;
- every page of a document longer than **25** pages with no printed page number anywhere (a footer template binding unrelated documents).

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9. The final pass: whole documents

After every gap is decided, one more pass looks at whole documents (`_final_pass`).

Split first. A document holding two or more **different dates read from a label** (trust 5) splits at each date change (`final_pass: split: two dates in one document`, flagged). This catches a document that spans two visits while every single gap looked like a join.

Then merge neighbours, but only across a **weak** break: confidence low or score ≤ 2, never a decisive break, and never between two documents with different dates. These run up to **3** times, because one merge can reveal another:

Rule	Merge two neighbouring documents when
`same_date`	both are dated and the dates are equal
`continued_across`	the first one's last page ends "continued"
`same_type_untitled`	both have the same page type (full path) and neither starts with a title
`orphan_page`	a one-page undated document sits between two documents of the same type, both breaks weak: it joins the one after
`junk_rejoin`	two junk groups of the same subtype

Last, a Progress Note runs on (`progress_note_runs_on`). A document whose first page is a Progress Note (`final_pass.progress_note_families`) keeps taking the documents after it until one of these ends it:

- a **new date read from a label** (trust 5) on the next document's first page;
- a **signature** on the note's last page (anywhere on it, or in its closing lines);
- a **decisive break** (a different labelled date, a run of three or more junk pages);
- a **junk group**.

So a lab table, a questionnaire or an untitled page after a note stays in the note. Unlike the merges above, this one also crosses a strong break such as a title.

A merged gap keeps its score; it records `final_pass: merged: <rule>` and the page's `decided_by` becomes `final_pass`. Each rule can be switched off in `final_pass.rules`.

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10. What comes out

10.1 Tags and labels

Break Before	Break After	Tag	Label
break	not	**new**	`New (Doc 3)`
not	not	**continue**	`Continue (Doc 3)`
not	break	**last**	`Last (Doc 3)`
break	break	**standalone**	`Standalone (Doc 3)`

Junk groups read `New (Junk 4)` and so on: documents and junk groups share **one numbering**, so `document_seq` carries both. A blank page reads `Blank (in Doc 3)` or `Blank`. The tag is derived from the document's pages after the final pass, never stored on its own. The legacy `position` column is still written (new → first, standalone → single).

10.2 `page_continuity_results`: the original columns

No new columns. One row per content or junk page:

Column	Holds
`document_seq`	the document or junk group (one numbering)
`seq`	page within it
`position`	the tag: `first` = New, `continue`, `last`, `single` = Standalone
`relation`	`new_document` for its first page, else `continue`
`score`	the **Break Before**
`decided_by`	the gap before: `chart_start`, `decisive`, `decisive_conflict`, `sum`, `tie`, `conflict_a`, `conflict_b`, `final_pass`; a transparent page `attached`
`confidence`	the gap before's confidence, 0–1 (`|2p − 1|` on the curve, §8); 1 on the chart's first page; added 2026-10-20 beside `score`
`confidence_level`	the gap before's (high / medium / low)
`evidence`	the gap before in words, e.g. `+8 break, high (signed_in_tail +3, last_numbered_page +3, has_title +3, signed_anywhere +2)`
`link_strength`	strong while every link back to the first page is a 5-point join (page numbers or the same labelled date)
`start_confirmed`	the first page has a title or prints "page 1"
`review_required`	§8.5

Blank and duplicate pages have no row. `imaging_final` gives a document's Final values to its **content** pages only: junk pages keep their own, whatever their `document_seq`.

10.3 `<chart>/staging/continuity.json`: every score

Written beside `extraction.json` and `blank_junk.json` on every run, and kept after the chart finishes:

```
        {"version": 1, "chart_name": "…", "written_at": "…", "canon_version": 4,
         "documents": [{"document": 1, "kind": "main", "pages": ["1.jpg", "2.jpg"],
                        "dos_from": "2024-03-28", "dos_to": "2024-03-28", "page_type": "Progress
Note"},
                       {"document": 2, "kind": "junk", "pages": ["6.jpg", "7.jpg"],
                        "dos_from": "Skipped", "dos_to": "Skipped"}],
         "pages": [{"page_name": "2.jpg", "kind": "main", "document": 1, "label": "Continue (Doc
1)",
                    "tag": "continue", "break_before": -8, "break_after": -8,
                    "decided_by": "decisive", "confidence": 0.96, "confidence_level": "high",
                    "review_required": false, "dos_from": "", "dos_trust": null, "printed": [2,
3],
                    "signed": false, "signed_in_tail": false, "title": false, "page_type": "…",
                    "visit_ids": [],
                    "gap_before": {"score": -8, "decision": "joined", "decided_by": "decisive",
                                   "confidence": 0.96, "level": "high",
                                   "ending": {}, "opening": {}, "pair": {"pagination_follows":
-5, "…": 0},
                                   "sub_scores": {"ending": 0, "opening": 0, "pair": -8},
                                   "recorded": {"text_start": -1}, "flags": [], "final_pass":
null},
                    "gap_after": {"…": "…"}}],
         "encounters": [{"seq": 1, "document": 1, "page_family": "Progress Note",
                         "pages": ["1.jpg", "2.jpg"], "dos_from": "2024-03-28", "dos_to": "2024-
03-28",
                         "encounter_type": "outpatient_f2f", "encounter_label": "Outpatient
(F2F)",
                         "confidence": 0.8, "decided_by": "tier2", "reason": "",
                         "matched_keyword": "office visit", "conflict": false, "contenders":
[]}]}
```


`encounters` is added at the end by the `encounter_type` stage (it runs after continuity): one entry per encounter (a continuity document), with the one decision every page of it carries. Review-ui's Doc Summary builds from it, and derives the encounters from `document_seq` as before when a chart has no staged `encounters`.

The CSV (`<chart>_continuity.csv`) lists every page (blank and duplicate too) with the label, both scores, the confidence, `kind`, `segment`, `dos_trust` and `layout_source`; a junk group's DOS reads `Skipped`.

10.4 What a document shares

Unchanged (`document_finals`, applied by `imaging_final`):

Final value	Rule
Page type	the document's first page's (each page keeps its own sub-type)
Date of service	the first dated page's (the first page's when it is dated), on every page
A stand-alone non-visit document	every page a `default_final_dos` type (demographics, registration, growth chart, problem list, immunizations): **2222-02-02**
Encounter type	one per document (the encounter stage reads `document_seq`)

Junk pages keep their own values; a junk group's DOS and encounter type read **Skipped** in review-ui's Doc Summary.

11. Worked example

An eight-page chart, as the engine scores it (the same pages are easy to rebuild in a test). Pages 1–3 are a progress note dated 2024-03-28 from a label and printed “page N of 3”, with one banner and footer; page 3 is signed in its closing lines and stops 60% down. Page 4 is a lab report with a title, stopping 40% down. Page 5 is blank. Pages 6–7 are a records request (junk, 96% sure). Page 8 is a note dated 2024-06-03.

Gap	Signals	Score	Decision
1 → 2	page numbers run on –5, same header –2, same footer –1, wording –0.6 / –0.9, cut mid-sentence –1, starts mid-thought –1	**–11.5**	joined, decisive
2 → 3	page numbers run on –5, same header –2, same footer –1	**–8**	joined, decisive
3 → 4	signed in the closing lines +3, signed +2, “page 3 of 3” +3, text ends in the band +2, page 4 has a title +3	**+13**	break, high
4 → 6 (blank 5 between)	a junk run of two starts +2.9 (3 × 0.96), blank between +1, page 4’s text ends at 40% +0.7	**+4.6**	break, medium: page 6 opens Junk 3
6 → 7	same junk kind –2, same header –2, wording –2 / –1, cut –1, mid-thought –1	**–9**	joined, high
7 → 8	the junk run ends +2.9	**+2.9**	break, medium

```
New (Doc 1)  Continue (Doc 1)  Last (Doc 1)     pages 1–3, 2024-03-28
Standalone (Doc 2)                              page 4, lab report
Blank                                           page 5, between documents
New (Junk 3)  Last (Junk 3)                     pages 6–7, records request
Standalone (Doc 4)                              page 8, 2024-06-03
```

Page 8’s date is not compared with anything: the document open before the junk run is Doc 2, which has no date. Had Doc 2 been dated 2024-03-28, the gap would add `different_dos` +5 (decisive). The final pass changes nothing here: every break is above the weak line (score ≤ 2 or low confidence).

12. Settings (`continuity_canon.json`, version 4)

Block	Holds
`points`	every signal: `side`, `points`, `scaled`, `decisive`, `status` (measured / assumed)
`dos_trust`	DOS tier → points (service / admit / discharge 5, weak / received / printed / collected 3, keyless 2, default 3)
`junk_run`	`decisive_at` 4.5
`page_fill`	the curve (92 / 80 / 65 / 50 / 35%) and the text proxy (28 lines / 1,800 characters)
`text_start`	the curve (3 / 12 / 25%), `apply` false
`semantic`	furniture 12 characters, 20-word blocks, n-grams, `min_df` 2, divisors 0.25 / 0.15
`section_order`	note order (chief complaint → … → follow-up) and the section names in each step
`decision`	high ≥ 5, medium ≥ 2, the curve's `a` 2 and `b` 0 (to fit), disagreement × 0.6, review below 0.25
`conflict_rules`	±4, `apply` false
`final_pass`	weak = score ≤ 2 or low confidence, 3 passes, each rule on / off, `progress_note_families`
`review_triggers`	long run 25 pages
`pagination`, `bands`, `default_final_dos`, `measurement`	as before

The loader refuses a file whose `points` name a signal the engine does not produce, or that leaves one out.

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13. What is proven, what is assumed, and what waits

Measured (170 labelled gaps from 15 charts, both pages printing page numbers): text reaching the bottom (34), same header line (31), wording with furniture removed (28, 27), sections in order (27, no false positives), signed (26) and signed at the bottom (21), same footer (8), same visit number (8, no false positives).

Assumed, to measure first: title (+3), page-type change, junk runs, "continued", same junk kind, the three decisive signals individually, the ending band, normalised header / footer matching.

Three things are measured and stored but do not change any answer yet. Each is one switch in `continuity_canon.json`:

1. **How far down the next page's text starts** (`text_start`, §5.4). The value is in `continuity.json` under `recorded`; it is not added to the score (`text_start.apply`).
2. **The two disagreement patterns** (§8.4). When they fire, the gap is flagged and its confidence drops one level, but the break / join answer stays the sum's (`conflict_rules.apply`).
3. **The 0–1 confidence number** (§8.3) uses two constants, `a` = 2 and `b` = 0, picked by hand. Fitting them on labelled gaps makes the number a better ranking; the decision itself does not use them.

Built for the final run only. The method allows re-scoring at earlier stages (after blank / junk, after page type) for audit; only the final score feeds anything, and that is the one the stage computes. Ink reaching the bottom of the image (an early-stage signal) is not used: by the final run the line positions are there.

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14. How to check it is working

- **Count the documents.** 623 pages making 9 documents was once a numbering bug.
- **Look at the longest document.** A long run with no page numbers is usually a footer template merging unrelated documents (the 25-page flag).
- **Plot document lengths.** A spike at one page means fragmenting; a long tail means merging.
- **Targets:** wrongly merged under 2% (the expensive error), low-confidence share under 30%, accuracy over 95% on the gaps whose answer is known.
- **Split validation by chart, never by page**, or the weights will look better than they are.